

Logistic regression model

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1 Model

We obtain n data points with d features.

We convert the output to values -1 and 1. That will be stored in $n \times 1$ matrix Y .

We may remove the reviews labeled neutral from our dataset (the data with known sentiment)

In module 2, d features means how often d -th word appears in the original review text

We insert a dummy column with ones as 0-th column which we'll map to the intercept w_0

So we put the features in $n \times (d+1)$ matrix H

We also have $d + 1$ coefficients which we put in a $(d+1) \times 1$ matrix w where w_0 is the intercept

The scores per data point are obtained by multiplying H with w :

$H \times w = \text{score}$ will be a $n \times 1$ matrix

After that we convert each single score to a probability using sigmoid function:

$$P = \frac{1}{1 + e^{-\text{score}}}$$

2 Side note

So far a data point was a row in an $n \times (d+1)$ matrix. But alternatively sometimes a row x_i is denoted as:

$$\begin{bmatrix} h_0(x_i) \\ h_1(x_i) \\ \vdots \\ h_d(x_i) \end{bmatrix}$$

Then the score of the i -th data point is subsequently written as dot product

$$score = w^\top h(x_i)$$